CAPSTONE 78 • AI FOR ACCESSIBILITY

Accessibility Begins With Access

Using AI tools to make learning materials easier to see, hear, and understand. Technology should remove barriers — not create new ones.

01
Tool Evaluation

02
Educator Guide

03
Python Workflow

PEOPLE & PURPOSE

One Campus. Different Needs. Equal Access.

01. Students with Disabilities
Visual, auditory, and cognitive learning aids.

02. Faculty & Educators
Simple workflows for accessible learning media.

03. Campus IT Teams
Software compliance and infrastructure support.

PROJECT OBJECTIVES

- ✓ **Benchmark:** Evaluate 4+ free AI tools for accuracy.
- ✓ **Compliance:** Establish WCAG 2.1 AA steps.
- ✓ **Automation:** Generate first-draft alt-text automatically.
- ✓ **Review:** Maintain mandatory human review workflows.

TOOLKIT

A Toolkit for Inclusive Learning

ChatGPT / Gemini

Draft image descriptions and structured alt-text.

ElevenLabs

Create natural-sounding text-to-speech audio files.

Live Captions

Transcribe lecture audio with real-time stream feedback.

NotebookLM

Summarize documents using W3C WAI source files.

Assistive Tech Integration

Screen Readers: NVDA & VoiceOver

Standard: W3C WCAG 2.1 AA

Access Modes: Audio, Text, Synchronized Captions

RESULTS

Evaluation Insights & Limitations

~90%
Alt-Text Accuracy
Standard educational images

85–90%
Transcription
Low-noise environments

High
Text-to-Speech
Reliable for academic prose

Limitations & Mitigation

Technical Jargon: Mathematical formulas require expert inspection. **Hallucinations:** Low-contrast diagrams may cause visual detail errors. **Human Verification:** Check every output before publication.

PRACTICE

Make Accessibility Part of the Workflow

Accessibility Checklist

- ☑ Meaningful alt-text for graphics
- ☑ Synchronized captions for video
- ☑ At least 4.5:1 contrast for body text
- ☑ Clear H1–H3 heading hierarchy

STAGE 01 • DRAFT
Python automation generates alt-text & caption drafts.

STAGE 02 • REVIEW
Educators verify context and factual accuracy.

STAGE 03 • PUBLISH
Multi-modal materials are shared via the LMS.

RESPONSIBLE AI

Keep People at the Centre

01. Lived Experience

Test tools directly with disabled users and integrate feedback.

02. Human Oversight

AI supports accessibility work — people make final judgments.

03. Data Privacy

Protect student information during model processing.

"AI can speed up captions and alt-text, but responsible accessibility requires human review. Universal design benefits every learner."